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Study of the Cox-Ross-Rubinstein model for option valuation

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1 Introduction

*"A good portfolio is more than a long list of
good stocks and bonds. It is a balanced whole [...]."*
– **H. Markowitz [1]** –

1.1 Motivation

This document was made to be an introduction to quantitative finance through development of market risk analysis algorithms. It focuses on the study of the binomial model, known as the Cox-Ross-Rubinstein model.

Firstly, we will talk about the theoretical foundations of this prediction model before analyzing how it differs from the Black-Scholes and Monte Carlo models. Finally, we will implement and analyze the market variation estimators, known as the "Greeks".

I am working on this paper in my spare time for personal learning and curiosity. Consequently, it may contain errors or inaccuracies. The provided sources are in French or English, you may use a translator if you do not understand either of these languages.

1.2 History of option valuation

Modern portfolio theory in finance originated in 1952 with the work published by Harry Markowitz [2]. This theory explains how investors can optimize their portfolios and analyze asset pricing based on market conditions (volatility, risk and so on). Therefore it lays the foundations of quantitative finance by defining the portfolio as a collection of financial assets.

Based on Markowitz's work, three economists *Fischer Black*, *Myron Scholes* and *Robert Merton* published an article in 1973 presenting a formula allowing to calculate the theoretical price of a European option. Their contribution was to find a partial differential equation (PDE) to find the fair price for a European call option. [3]

With the development of computers and the exponential growth of computing power, a new method emerged in 1977 : the Monte Carlo approach. This model involves randomly simulating thousands of possible price paths. The option price is the the average of the payoffs obtained from these paths. [4] It is the primary method for exotic options such as Asian options, where the payoff depends on the average price over a certain period of time.[5]

Two years later, a new trio of economists : *John Cox*, *Stephen Ross* and *Mark Rubinstein* proposed an innovative alternative : the binomial model, also known as the Cox-Ross-Rubinstein model. In this model, time is discretized, allowing the path of an asset price to be represented as a binary tree. This simplifies estimation, providing a numerical approximation to the Black-Scholes model which relies on an analytical solution. The approach shifts the problem from a complex partial differential equation to a simple approximation that can be programmed on a computer. Furthermore, the model enables the valuation of American options, which can be exercised at any time with only a minor change in the algorithm. [6]

Subsequently, other more exotic models appeared (e.g. trinomial trees) [7]. In one of the following part, we will focus on the binomial model and compare it with the Black-Scholes and Monte Carlo models, representing respectively an approximation model, an analytical model

and a simulation-based one. The Cox-Ross-Rubinstein model is more intuitive and simpler to implement algorithmically than the Black-Scholes model.

1.3 Goals

The main goal of this paper is to present the theoretical foundations of the binomial model, implement it and then compare it to other option pricing models.

We will begin by establishing the basic terminology and presenting key theoretical results from financial mathematics. Once the model's algorithmic implementation is complete, we will explore its capabilities by pricing complex options. We will analyze how the model's discrete structure allows for the valuation of products that cannot be handled analytically by the Black-Scholes model, such as American options which can be exercised early. The study will be complemented by the calculation of sensitivity indicators known as *Greeks*, which are essential risk management tools for measuring a portfolio's sensitivity to market volatility.

Finally, we will compare our results with those obtained from the Black-Scholes and Monte Carlo models. This comparative analysis will enable us to highlight the binomial model's convergence rate and identify its limitations when applied to real-world data, thereby preparing for potential extensions involving more sophisticated tree structures.

1.4 Vocabulary

Before proceeding with the study of quantitative finance and the algorithms behind the Cox-Ross-Rubinstein model, it is essential to establish a basic vocabulary necessary for a proper understanding of the remainder of the document. If you have a financial background, you can skip this part as it will only present the basic key ideas in market finance.

Derivative : A derivative is a type of financial contract whose value depends on an asset known as **underlying asset** and whose price fluctuates in line with changes in the rate or price of that underlying asset. There are three main categories of derivatives : **firm commitment products**, **swaps** and **contingent products or options** [8] [9] [10]

Forward product : This group of derivative products generally includes **forward** and **futures**. A **forward** contract is a financial agreement between two entities, agreeing on a defined sale and purchase price at a fixed deadline. Both entities must honor their parts of the contract even if the situation at maturity is unfavorable to them. There are some differences between **forwards** and **futures** : forward contracts are negotiated over the counter, non-standardized and not very negotiable, whereas futures contracts are negotiated on a stock exchange, are standardized and negotiable. [8] [9] [11] [12]

Swap : Exchange of financial flows between two entities (e.g. floating-rate interest for fixed-rate interest). [12]

Option : Establishes a right between a buyer and a seller to buy (**Call**) or sell (**Put**) an underlying asset at a predetermined price (**Strike**) by a set expiration date. This derivative represents a right and is not an obligation. The **Call/Put** symmetry allows one to anticipate either a favorable movement in the underlying asset's price (Call) or an unfavorable one (Put). There are numerous types of options, notably **European** options that are exercisable only at expiration, and American options that are exercisable at any time prior to expiration. These represent the simplest forms, known as Vanilla options. [13] [3] The **Payoff** refers to the outcome at expiration.

An option is :

- **In the money / ITM** when its strike price is lower than the price of its underlying asset for a call, or higher than the price of its underlying asset for a put.
- **Out of the money / OTM** when its strike price is higher than the price of its underlying asset for a call, or lower than the price of its underlying asset for a put.
- **At the money / ATM** when its strike is equal to the current price of the underlying asset.

We will use the following notations :

- K : Strike
- S : Price of the underlying asset
- p : Option Premium
- R : Profit at maturity
- P : Payoff

Payoff Formulas

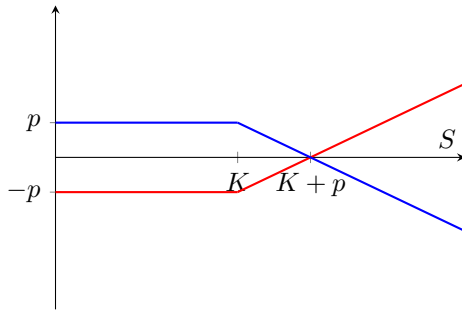
$$P_{Call} = \max(S - K, 0)$$

$$P_{Put} = \max(K - S, 0)$$

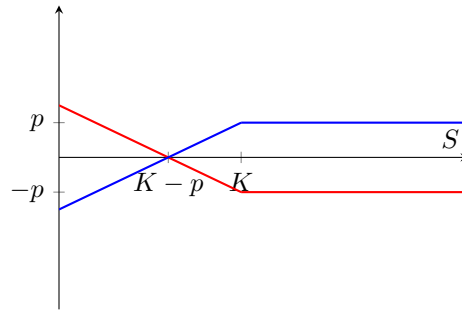
We can represent profits at maturity :

$$R = \begin{cases} R_{\text{Call} - \text{Buyer}} = \max(S - K, 0) - p \\ R_{\text{Call} - \text{Seller}} = -\max(S - K, 0) + p \end{cases}$$

$$R = \begin{cases} R_{\text{Put} - \text{Buyer}} = \max(K - S, 0) - p \\ R_{\text{Put} - \text{Seller}} = -\max(K - S, 0) + p \end{cases}$$



(a) Option - Call



(b) Option - Put

Figure 1: Profit of Call and Put options

Options are a key concept to understand quantitative finance. It is useful to work through a numerical example to see how it works :

Consider an industrial buyer of wheat, **B**, who knows on **January 1st** that they will need to purchase $n = 1\,000$ tonnes of wheat on **June 30th**. On January 1st, the price of wheat is 200\$ per tonne. B anticipates that the price of wheat will rise. Since B would begin to lose money if the price exceeds 250\$, they decide to purchase 1 000 call options with a strike price of 250\$, a premium of $p = 10$ \$ per tonne and an expiration date of June 30th.

Several scenarios are possible for June 30th :

- **Case 1:** Wheat trades at 150\$ per tonne. The call option is **OTM**. B's prediction did not materialize. The call option is now worthless, leading B to let it expire. B thus loses $n \cdot p = 1000 \cdot 10 = 10\,000$ \$. B therefore purchases the wheat on the market at 150\$ per tonne, resulting in a total cost of $150 + p = 160$ \$ per tonne.
- **Case 2:** Wheat trades at 225\$ per tonne. The call option is **OTM**. B's prediction has partly come true but not enough to make the call options worth using. The call option is still worthless, leading B to let it expire. B thus loses $n \cdot p = 1000 \cdot 10 = 10\,000$ \$. B therefore purchases the wheat on the market at 225\$ per tonne, resulting in a total cost of $225 + p = 235$ \$ per tonne.
- **Case 3:** Wheat trades at 250\$ per tonne. The call option is **ATM**. B's prediction has partly come true but not enough to make the call options worth using because the price is equal to the strike. Using the call option does not change anything. B thus loses $n \cdot p = 1000 \cdot 10 = 10\,000$ \$. B therefore purchases the wheat on the market at 250\$ per tonne, resulting in a total cost of $250 + p = 260$ \$ per tonne.
- **Case 4:** Wheat trades at 300\$ per tonne. The call option is **ITM**. B's prediction came true. B uses its call options and spend 250\$ per wheat tonne. Thus, B will have spent $250 + p = 260$ \$ per tonne instead of 300\$. B saved $(300 - (250 + p)) \cdot n = 40 \cdot 1000 = 40\,000$ \$.

Therefore, the goal of market finance is to optimize strategies in order to generate the highest possible profit while taking risk into account. Thus, it is useful to develop mathematical tools that allow for the modeling of market dynamics.

Positions : There are two positions when buying or selling options :

- **Long Position**

Action Buying the asset in anticipation of a price increase

Profit Making money by reselling at a higher price than the purchase price

Risk Risk limited to the amount invested

- **Short Position**

Action Borrowing an asset from someone, selling it immediately, waiting for the price to drop and buying back the asset at the lower price to return it

Profit Gaining the difference

Risk Theoretically infinite risk if the share price rises without limit

2 Mathematical theory and no-arbitrage principle

In this section, we will focus on the theoretical notions and concepts necessary to understand finance.

2.1 Filtered probability space

Consider the triplet $(\Omega, \mathcal{F}, \mathbb{P})$ forming a probability space, with :

- the sample space Ω representing the possible states of a financial market
- a σ -algebra $\mathcal{F}$ on the sample space Ω
- a probability measure $\mathbb{P}$ on the σ -algebra $\mathcal{F}$

Let $I = \{0, 1, \dots, T\}$, with $T < \infty$ be the discrete set representing time points, where T is the option's maturity date. We introduce a **filtration** $(\mathcal{F}_t)_{t \in I}$ on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The filtration $(\mathcal{F}_t)_{t \in I}$ is an increasing set of sub- σ -algebra of $\mathcal{F}$ with the property :

$$\forall (t_0, t_1) \in I \times I, \quad t_0 \leq t_1 \implies \mathcal{F}_{t_0} \subseteq \mathcal{F}_{t_1} \subseteq \mathcal{F}$$

[14] We then define the **filtered probability space** $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in I}, \mathbb{P})$

Remark. Working within this space allows us to model the evolution of a financial market. Thus, each σ -algebra in the filtration $(\mathcal{F}_t)_{t \in I}$ represents the information known at time t . Therefore, an agent at time t_1 has access to all information from an earlier time t_0 , provided $t_0 \leq t_1$. The studied stochastic process $X := (X_t)_{t \in I}$ must be adapted to the filtration $(\mathcal{F}_t)_{t \in I}$ ($\forall t \in I$, X_t is $\mathcal{F}_t$ -measurable). We can now define the random variables used to model the financial processes of the model.

Consider the following random variables :

$$\begin{aligned} \text{Asset quantity :} \quad & \alpha_t : \Omega \rightarrow \mathbb{R} \\ \text{Risky asset :} \quad & S_t : \Omega \rightarrow \mathbb{R}^+ \\ \text{Risk-free asset :} \quad & B_t : \Omega \rightarrow \mathbb{R}^+, t \mapsto (1 + r)^t \end{aligned}$$

with $r > -1$ the risk-free rate. We assume that the random variables α_t , S_t and B_t are $\mathcal{F}_t$ -measurable for all $t \in I$.

Remark. For each $t \in I$, the random variable B_t is constant on Ω ($\forall \omega \in \Omega, B_t(\omega) = (1 + r)^t$). Therefore, it is a deterministic random variable. S_t represents financial market uncertainty, α_t is a strategy adapted to the filtration (and thus adapted to the information known at time t), and B_t represents the deterministic evolution of a risk-free investment.

2.2 Portfolio theory

We then introduce the concept of an **investment strategy** as a stochastic process $\Theta = (\Theta_t)_{t \in I} := (\alpha_t, \beta_t)_{t \in I}$ where α_t and β_t represent respectively the number of units at time t in the risky asset and in the risk-free asset. This stochastic process is adapted to the filtration $(\mathcal{F}_t)_{t \in I}$.

Let $B_t = (1 + r)^t$ be the value of one unit invested in the risk-free asset. Thus, the value of the risk-free account is given by $\beta_t B_t$.

A **discrete portfolio** Π_t at time t is defined by the value :

$$\Pi_t = \alpha_t S_t + \beta_t B_t$$

Therefore, the stochastic process Π_t is composed of two parts : $\alpha_t S_t$ representing the risky component of the portfolio, and $\beta_t B_t$ representing the risk-free component of the account. [15]

Although the model evolves in discrete time, each transition from t to $t + 1$ contains two phases :

- Price Update : Asset prices change from (S_t, B_t) to (S_{t+1}, B_{t+1}) . The value of the portfolio held from period t , prior to any updates, becomes $\alpha_t S_{t+1} + \beta_t B_{t+1}$.
- Portfolio update : The investor updates their holdings from (α_t, β_t) to $(\alpha_{t+1}, \beta_{t+1})$ at the newly observed prices (S_{t+1}, B_{t+1}) . The value of the new portfolio is $\alpha_{t+1} S_{t+1} + \beta_{t+1} B_{t+1}$.

Self-financing property

A discrete portfolio is said to be **self-financing** if

$$\forall t \in I, \quad \Pi_{t+1} = \alpha_t S_{t+1} + \beta_t B_{t+1} = \alpha_{t+1} S_{t+1} + \beta_{t+1} B_{t+1}$$

[16]

Thus, the self-financing property ensures that the evolution of Π_t is entirely determined by the strategy Θ and the dynamics of both S_t and B_t , without external adjustments. In the following, we will assume that all portfolios are self-financing.

No-Arbitrage Principle

The No-Arbitrage principle means that there is no strategy Θ ensuring a self-financing portfolio with zero initial value and positive final value almost surely.

$$\nexists \Theta, \Pi_0 = 0, \mathbb{P}(\Pi_T \geq 0) = 1, \mathbb{P}(\Pi_T > 0) > 0$$

Remark. The No-arbitrage principle forbid the theoretical existence of a risk-free asset requiring no initial investment. Consequently, any potential gain requires an initial outlay or entails risk. Without this condition, every investors would exploit arbitrage opportunities, causing markets to collapse. This property is a necessary condition to ensure market consistency. In practice, very minor arbitrage opportunities appear during market fluctuations and are exploited by arbitrageurs. [17]

2.3 Replication strategy

Replication

A payoff H is said to be **replicable** if : $\exists \Pi, \Pi_T = H \quad \mathbb{P}$ -a.s.
 Π is called the **replicating portfolio** for H .

Market Completeness

A market is said to be **complete** if every payoff is replicable with a portfolio.

Remark. It follows naturally that market completeness implies the existence of a replicating portfolio for every payoff. However, one can go further and demonstrate the uniqueness of the portfolio for each payoff.

Consider a complete market, for any payoff H , there exists a **unique** replicating portfolio for H .

Proof. Let $\Pi = \Pi^1 - \Pi^2$ be the difference between two self-financing portfolios that both replicate the payoff H . Since the difference of two self-financing portfolios is self-financing, Π is self-financing. Furthermore,

$$\Pi_T = \Pi_T^1 - \Pi_T^2 = H - H = 0$$

We will now prove that $\Pi_t = 0$ for every $t \in [0, T]$.

Suppose by contradiction that there exists $t \in [0, T]$ such that $\Pi_t \neq 0$.

If $\Pi_t > 0$, take the opposite position $-\Pi$ at time t . This creates an initial cash amount $\Pi_t > 0$. Invest this amount in the risk-free asset until time T .

Since $\Pi_T = 0$, the position $-\Pi$ has zero final value, while the amount invested in the risk-free asset has strictly positive value

$$\Pi_t \frac{B_T}{B_t} > 0$$

Thus, starting from zero initial wealth at time t , we obtain a strictly positive final payoff, yielding an arbitrage opportunity.

If $\Pi_t < 0$, take the position Π at time t and borrow $-\Pi_t > 0$ from the risk-free asset. The initial value of the combined position is $\Pi_t + (-\Pi_t) = 0$.

At time T , the portfolio Π has value $\Pi_T = 0$, while the loan has to be repaid in the amount

$$(-\Pi_t) \frac{B_T}{B_t} > 0$$

Therefore this position produces a strictly negative final payoff. Taking the opposite position produces an arbitrage opportunity. In both cases, $\Pi_t \neq 0$ leads to an arbitrage opportunity, contradicting the no-arbitrage assumption.

Therefore, $\Pi_t = 0$ for all $t \in [0, T]$

Consequently, $\Pi_t^1 = \Pi_t^2$ for all $t \in [0, T]$

Thus the self-financing replicating portfolio for H is unique. $\square$

2.4 Risk-neutral measure

Fundamental Theorems of Asset Pricing

First Theorem : A discret financial market on the filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in I}, \mathbb{P})$ is arbitrage-free if and only if there exists at least one risk neutral probability measure that is equivalent to the original probability measure $\mathbb{P}$.

Second Theorem : An arbitrage-free market is complete if and only if there exists a unique risk-neutral measure, equivalent to $\mathbb{P}$ for which the **numeraire** (reference asset used to express prices) is the risk-free asset.

[18] [19] [20]

Remark. We previously required markets to satisfy the no-arbitrage principle. Thus, we assume the existence of a risk-neutral measure. Furthermore, if the market is complete, the risk-neutral measure is **unique**, allowing for the consistent pricing of all assets. This theorem forms the basis of models for valuing options and derivatives.

The second theorem states that a market is complete if and only if there exists a unique risk-neutral probability measure. In the binomial model, this property corresponds to the fact that the number of future states (up/down) equals the number of assets (risky/risk-free). Thus, the model considered so far involves two types of assets (risky/risk-free), and only two possible scenarios (rise/fall) can be considered at each step. This justifies the use of binomial models, which is the easiest complete model, as opposed to a trinomial model, which would be incomplete.

Martingale

$M := (M_t)_{t \in I}$ is an $\mathcal{F}$ -martingale if :

- $\forall t \in I, M_t$ is $\mathcal{F}_t$ -measurable
- $\forall t \in I, M_t$ is integrable
- $\forall s \leq t, M_s = \mathbb{E}(M_t | \mathcal{F}_s)$ a.s.

Thus, the concept of martingale is useful to represent a financial market. Intuitively, the future expectation of a martingale, taking the past into account, is equal to the current value ($\mathbb{E}(M_{t+1} | \mathcal{F}_t) = M_t$).

Consider a discret market with a risk asset S_t and a risk-free asset B_t . The **discounted cash flow** of the risky asset is defined as :

$$\tilde{S}_t := \frac{S_t}{B_t}$$

[21]

We obviously see that the price is expressed in terms of B_t , which thus becomes the numeraire. According to the first fundamental theorem of asset pricing, if the market is **arbitrage-free**, there exists at least one measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ such that :

$$\tilde{S}_t = \mathbb{E}_{\mathbb{Q}}[\tilde{S}_{t+1} \mid \mathcal{F}_t]$$

In other words, under $\mathbb{Q}$, the discounted price becomes an $\mathcal{F}$ -martingale.

To better understand this concept, let's look at an example : Consider a market with :

$$S_0 = 100, \quad S_1 = \begin{cases} 120 \text{ with probability } p \\ 90 \text{ with probability } 1 - p \end{cases} \quad \text{avec } r_f = 5\%$$

We search the risk-neutral probability q such that the discounted cash flow is a martingale under $\mathbb{Q}$:

$$\tilde{S}_0 = \frac{1}{1+r} \mathbb{E}_{\mathbb{Q}}[S_1 \mid \mathcal{F}_0] = \frac{120q + 90(1-p)}{1+r_f}$$

The solution q corresponds to the **risk-neutral probability** of an asset price increase. Under this probability, the discounted cash flow $\tilde{S}_t$ is a martingale and is used to value all derivative products.

Let Π_T be the **payoff** of an asset at maturity T . If the market is **arbitrage-free**, the first fundamental theorem guarantees the existence of a risk-neutral measure $\mathbb{Q}$.

The initial price Π_0 of the asset is defined as the discounted expectation of its payoff under the measure $\mathbb{Q}$:

$$\Pi_0 = \mathbb{E}_{\mathbb{Q}}\left[\frac{\Pi_T}{B_T}\right] = \frac{\mathbb{E}_{\mathbb{Q}}[\Pi_T]}{B_T} = \frac{\mathbb{E}_{\mathbb{Q}}[\Pi_T]}{(1+r)^T}$$

This equality holds because B is deterministic.

Example: Consider a European call option with strike $K = 100$, maturity $T = 1$ period, and underlying asset S_T such that:

$$S_T = \begin{cases} 120 \text{ with probability } p \\ 50 \text{ with probability } 1 - p \end{cases} \quad \text{avec } r_f = 5\%.$$

The option being a call, its payoff is $\Pi_T = \max(S_T - K, 0)$.

The initial price is :

$$\Pi_0 = \frac{\mathbb{E}_{\mathbb{Q}}[\Pi_T]}{1+r_f} = \frac{1}{1+r_f} (20q + (1-q) \cdot 0) = \frac{20q}{1+r_f}$$

q is the risk-neutral probability of an increase in the asset's price. Under this probability, the discounted option price is consistent with an arbitrage-free market.

3 Analysis of the Cox-Ross-Rubinstein model

In this section, we will examine the theory behind the Cox-Ross-Rubinstein model to implement it later in the next section.

3.1 Objectives and Structure of the Model

The objective of the **binomial** or **Cox-Ross-Rubinstein** model is to find the no-arbitrage price of the derivative product. The **no-arbitrage price** is the price at which no arbitrage opportunity exists. This model is based on portfolio replication, the absence of arbitrage and market completeness to exactly replicate the option's future cash flows and to guarantee the existence and uniqueness of a risk-neutral probability measure.

The **Cox-Ross-Rubinstein** model is a discrete pricing model unlike the **Black-Scholes** model which is continuous. The main advantages of working in discrete time are the ease of modeling and algorithmic implementation. Indeed, the binomial model is intuitively easy to understand as it represents asset evolution along the branches of a binary tree weighted by the probabilities of price increases and decreases. We will focus on this representation in the remainder of this section.

To discretize the time horizon of the call, we consider subdivision of the option's validity period into N periods of equal length, yielding the following divisions :

$$t_n = \frac{nT}{N}, \forall n \in 0, \dots, N$$

Remark. Letting N tend towards very large values amounts to approaching a continuous model.

For each subdivision, two factors are defined :

- u** : the asset's upward movement factor (up)
- d** : the asset's downward movement factor (down)

These two factors correspond to the rise and fall in the asset's price, respectively. In a tree diagram, we will consider the convention of placing the up-factor on the upper branch and vice-versa.

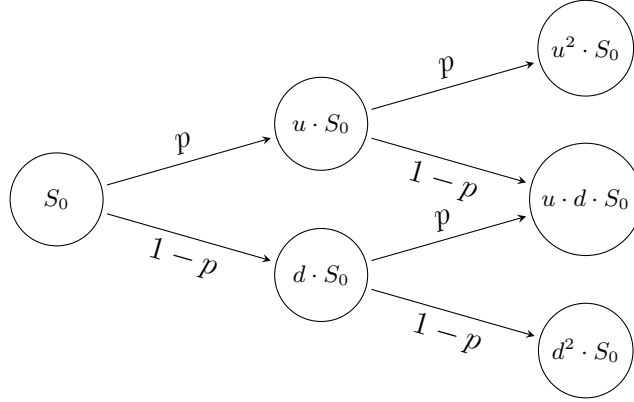
Remark. For the time being, we won't dwell on how these two factors are calculated.

The algorithm therefore breaks down into two distinct phases :

- **The propagation phase (Forward Phase)** : Starting from S_0 the underlying asset price tree is constructed up to maturity T .
- **The evaluation phase (Backward Phase)** : Once the prices are set at maturity, the **payoffs** on the tree leaves are calculated. Then, we go backwards in the tree traversal to determine the payoff S_0

3.2 Forward Phase

Let us represent the price evolution of an underlying asset with an initial price S_0 , which has a probability p of increasing by u and a probability $1 - p$ of decreasing by d in each period:



Remark. The tree is recombining; price paths are invariant under permutations of the up and down factors. More formally, let $w = (w_1, w_2, \dots, w_n)$ be a word of length n over the alphabet $\{u, d\}$. If we denote $|w|_u = k$ as the number of up moves and $|w|_d = n - k$ as the number of down moves, then for any permutation $\sigma \in \mathcal{S}_n$, the final price is identical:

$$S_n = S_0 \cdot \prod_{i=1}^n w_i = S_0 \cdot \prod_{i=1}^n w_{\sigma(i)} = S_0 u^k d^{n-k}$$

This highlights the advantage of discretization. Indeed, for a non-recombining binary tree, the $N = 10$ stage would be represented by $2^{10} = 1024$ leaves (terminal nodes), whereas for a recombining tree, the number of leaves is $N + 1$. In short, tree recombination transforms the exponential growth in the number of leaves into linear growth.

The concept behind the Cox-Ross-Rubinstein model (and the Black-Scholes model) is to use portfolio replication to mimic the evolution of a derivative product. Since we have assumed the absence of arbitrage opportunities (AOA), the market is complete, and thus any payoff can be replicated, as previously discussed. Let us model the evolution of a portfolio to establish relationships between the model's various coefficients. By convention, we will use the *Continuous Compounding Formula* for the discount factor : $e^{-r\Delta t}$ with $\Delta t := \frac{T}{N}$. We could also use the discret compounding one $\frac{1}{1+r}$ but it is less precise and it will be easier to prove the convergence to the Black-Scholes model using the continuous one. Let $\Pi = \Delta S + \beta B$, where B earns the risk-free rate r . We define the risk-free factor $R := 1 + r$. Let u and d be the model's evolution factors, and p the probability of an upward movement. We consider a call option C whose price can rise (C_u) or fall (C_d), and we construct our model to replicate this call.

$$\begin{array}{ccc}
& & \Delta uS + \beta B e^{r\Delta t} \\
& \nearrow \mathfrak{p} & \\
\Delta S + \beta B & & \\
& \searrow I - \mathfrak{p} & \\
& & \Delta dS + \beta B e^{r\Delta t}
\end{array}$$

Thus, since the portfolio replicates C , we obtain:

$$\begin{cases} \Delta uS + \beta B e^{r\Delta t} = C_u \\ \Delta dS + \beta B e^{r\Delta t} = C_d \end{cases}$$

Thus,

$$\begin{aligned}
\begin{cases} \Delta uS + \beta B e^{r\Delta t} = C_u \\ \Delta dS + \beta B e^{r\Delta t} = C_d \end{cases} &\iff \begin{cases} -\Delta u dS - d\beta B e^{r\Delta t} = -dC_u \\ \Delta u dS + u\beta B e^{r\Delta t} = uC_d \end{cases} \\
&\implies \beta e^{r\Delta t} B(u - d) = uC_d - dC_u \implies \beta B = \frac{uC_d - dC_u}{e^{r\Delta t}(u - d)}
\end{aligned}$$

With substitution,

$$\begin{aligned}
C = \Delta S + \beta B &= \frac{C_u - C_d}{S(u - d)} S + \frac{uC_d - dC_u}{e^{r\Delta t}(u - d)} \\
&= \frac{e^{r\Delta t}(C_u - C_d) + uC_d - dC_u}{e^{r\Delta t}(u - d)} \\
&= e^{-r\Delta t} \left(\frac{e^{r\Delta t} - d}{u - d} C_u + \frac{u - e^{r\Delta t}}{u - d} C_d \right) \\
&= e^{-r\Delta t} (pC_u + (1 - p)C_d), \text{ avec } p = \frac{e^{r\Delta t} - d}{u - d} \text{ et } 1 - p = \frac{u - e^{r\Delta t}}{u - d}
\end{aligned}$$

[21] **Remark.** Indeed, $1 = p + (1 - p) = \frac{e^{r\Delta t} - d + u - e^{r\Delta t}}{u - d} = 1$. We also necessarily have $d < e^{r\Delta t} < u$

Indeed,

If $e^{r\Delta t} \leq d < u$: One borrows at the risk-free rate

- In the event of a drop: $dS_0 - e^{r\Delta t}S_0 = S_0(d - e^{r\Delta t}) \geq 0$
- In the event of a rise: $uS_0 - e^{r\Delta t}S_0 = S_0(u - e^{r\Delta t}) \geq 0$

If $d < u \leq e^{r\Delta t}$: One sells short

- In the event of a rise: $e^{r\Delta t}S_0 - uS_0 = S_0(e^{r\Delta t} - u) \geq 0$

- In the event of a drop: $e^{r\Delta t}S_0 - dS_0 = S_0(e^{r\Delta t} - d) \geq 0$

In all four cases, an arbitrage opportunity arises, contradicting the No-Arbitrage Assumption. Thus, the risk-free return lies strictly between the best and worst risky returns.

The formula can be rewritten using the risk-neutral probability.

Risk-neutral valuation formula

Let u and d be the evolution factors with $d < e^{r\Delta t} < u$, where $e^{r\Delta t}$ is the risk-free factor, and let C_u and C_d be the future option payoffs:

$$C_0 = \frac{1}{e^{r\Delta t}}(pC_u + (1-p)C_d), \text{ with } p = \frac{e^{r\Delta t} - d}{u - d} \text{ and } 1 - p = \frac{u - e^{r\Delta t}}{u - d}$$

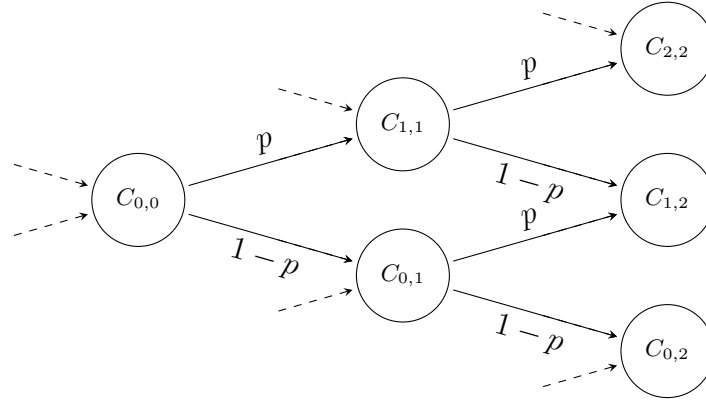
$$= \frac{\mathbb{E}^{\mathbb{Q}}(C_1)}{e^{r\Delta t}}, \text{ with } \mathbb{Q} \text{ being the risk-neutral probability measure}$$

3.3 Backward Phase

We can now apply the principle known as **backward induction**. This involves starting at the end of the tree and working back to the preceding node to calculate its value—defined as the expected value of the subsequent nodes—and repeating this process iteratively until the root is reached. Due to the non-linear nature of the valuation formulas for calls and puts (involving the *max* function), the values of the terminal nodes can only be determined once the tree construction is complete. Consequently, the tree must be built in its entirety before the backward induction step is performed. This principle is also highly useful for handling **American options** and for **delta hedging**.

Notons $C_{i,t}$ les noeuds dans un arbre binomial, où i est l'indice du noeud représentant le nombre de montées et t la couche de l'arbre représentant le temps.

Considérons alors le morceau d'arbre suivant :



Using backward induction, the value of node $C_{0,1}$ can be expressed in terms of the values of $C_{0,2}$ and $C_{1,2}$. We thus obtain:

$$C_{0,1} = e^{-r\Delta t}(pC_{1,2} + (1-p)C_{0,2})$$

Thus, we can work backwards up the tree from the leaves to evaluate all preceding nodes:

$$C_{i,t} = e^{-r\Delta t}(pC_{i+1,t+1} + (1-p)C_{i,t+1})$$

where i represents the position of the node among the nodes at depth t . The values at the leaves are determined by the call (or put) formula, depending on whether a call (or put) is being modeled. In this case, we have: $C_{i,N} = \max(S_{i,N} - K, 0)$, where S is the price of the underlying asset and K is the strike price.

Let us apply the binomial model to an option in the following example:

Data:

- Current price: $S_0 = 100$
- Strike price: $K = \$100$
- Up factor: $u = 1.1$
- Down factor: $d = 0.9$
- Risk-free rate: $r = 0.05$, so $R = 1.05$
- Number of periods: $N = 2$

Downward phase:

We calculate the possible paths

Time t=1:

- $S_u = 100 \times 1.1 = 110$
- $S_d = 100 \times 0.9 = 90$

Time t=2:

- $S_{uu} = 110 \times 1.1 = 121$
- $S_{ud} = 110 \times 0.9 = 99$
- $S_{dd} = 90 \times 0.9 = 81$

Calculation of risk-neutral probabilities:

$$p = \frac{R - d}{u - d} = \frac{1.05 - 0.9}{1.1 - 0.9} = \frac{0.15}{0.20} = 0.75$$

Calculation of leaf values:

We calculate the call option payoff:

- $C_{uu} = \max(121 - 100, 0) = 21$
- $C_{ud} = \max(99 - 100, 0) = 0$
- $C_{dd} = \max(81 - 100, 0) = 0$

Backward induction phase:

Time $t = 1$:

$$C_u = \frac{1}{1.05}(0.75 \cdot 21 + 0.25 \cdot 0) = \frac{15.75}{1.05} = 15$$
$$C_d = \frac{1}{1.05}(0.75 \cdot 0 + 0.25 \cdot 0) = 0$$

Time $t = 0$:

$$C_0 = \frac{1}{1.05}(0.75 \cdot 15 + 0.25 \cdot 0) = \frac{11.25}{1.05} \approx 10.71$$

For this option, the fair price to charge is \$10.71.

3.4 American Options

So far, we have focused on modeling the price of so-called European options—that is, options that can only be exercised at the expiration date. The strength of the binomial model lies in the ease with which it can be adapted to American options. Unlike a European option, an American option can be exercised at any time, rather than only at expiration. One could also examine Bermudan options, which are exercisable only on specific dates prior to expiration; however, for the sake of simplicity, we will limit our study in this section to American options. Due to this flexibility, an American option is worth at least as much as a European option. Indeed, an American option offers all the features of a European option but allows for earlier exercise, thereby making it possible to avoid a significant drop in value—for instance, in the case of an American call option.

Assuming an investor is rational—meaning they will always choose the decision that maximizes profitability—adapting our model becomes straightforward. One simply takes the maximum of the model-derived value and the exercise value when working backward through the tree. Consequently, American and European call options will have the same value, whereas an American put option will be more expensive than a European one, because the holder of an American put can exercise the option to prevent the price from falling below a certain level.

European formula: We simply discount the expected value of the next period.

$$C = e^{-r\Delta t}[pC_u + (1 - p)C_d]$$

American formula: At each node, we compare this discounted value with the payoff from immediate exercise and take the maximum of the two.

$$C = \max(\text{Intrinsic Value}, e^{-r\Delta t}[pC_u + (1 - p)C_d])$$

4 Algorithm Implementation

TODO

5 Convergence towards the Black-Scholes model

Before continuing our study of the binomial model, it is necessary to understand that the Cox-Ross-Rubinstein model is a discretization of another model that we will examine in this section: the **Black-Scholes** model.

5.1 Brownian motion / Wiener process

Let us begin by introducing a highly useful mathematical phenomenon: **Brownian motion**, first described in 1827 by the botanist **Robert Brown**. Brownian motion characterizes the random movement of a particle within a fluid as it went through collisions with other particles in the medium. Between collisions, the particle moves in a straight line at a constant velocity. The nature of this chaotic process can be modeled using a stochastic process known as the **Wiener process**, which is applicable to representing uncertain market dynamics. In 1933, the mathematician Paul Lévy demonstrated that Brownian motion is a continuous martingale, a property that allows us to apply the results discussed in previous sections. We denote by $W_t : \mathbb{R}^+ \rightarrow \mathbb{R}^d$ the Wiener process associated with Brownian motion at time $t \in I$ in a d -dimensional space. [22]

Wiener process

We say that a family of real random variables $(W_t)_{t \in \mathbb{R}^+}$ is a Wiener process if:

- $W_0 = 0$
- $\forall \omega \in \Omega, t \mapsto W_t(\omega)$ is continuous
- $\forall t_0 < t_1 < \dots < t_n, \forall i \in \{1, \dots, n\}, W_0, W_{t_1} - W_{t_0}, \dots, W_{t_n} - W_{t_{n-1}}$ are independent

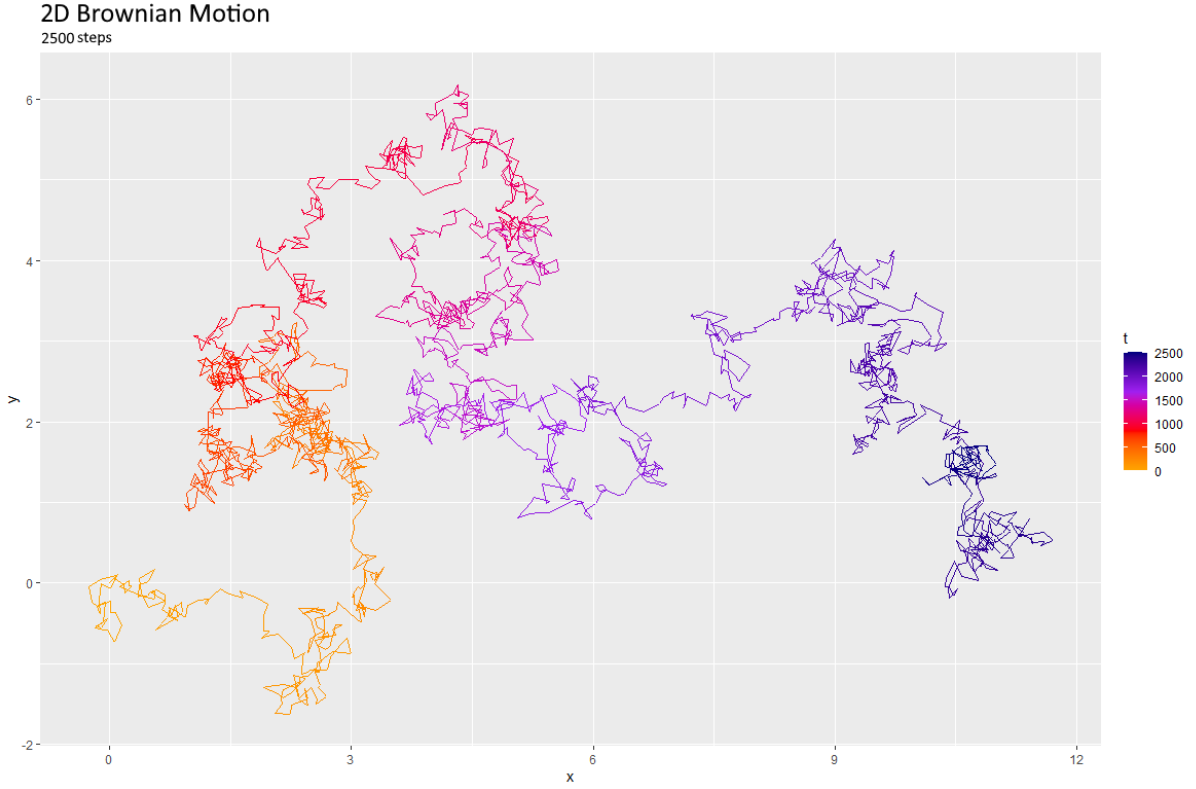


Figure 2: Simulation of a 2D Brownian motion

Remark. A Wiener process is continuous everywhere but differentiable nowhere (almost surely).

Remark. Le processus de Wiener est continu partout mais dérivable nulle part. It can be compared to a Weierstrass function or a fractal figure.

Considering the displacement of a particle in a d -dimensional space, we can associate the evolution of its position with the Wiener process $W_t : \mathbb{R}^+ \rightarrow \mathbb{R}^d$. Due to the nature of Brownian motion, the particle's mean position at time t is $\mathbb{E}[W_t] = 0$, since Brownian motion has no preferred direction. Thus, we immediately observe that $\text{Var}(W_t) = \mathbb{E}[W_t^2] - \mathbb{E}[W_t]^2 = \mathbb{E}[W_t^2]$.

In the following part, we will consider only one-dimensional Brownian motion.

It can also be shown that W_t is a special case of a Gaussian stochastic process:

$$\forall A \subset I, \forall (a_t)_{t \in A} \subset \mathbb{R}, \sum_{s \in A} a_s W_s \sim \mathcal{N}(\mu, \sigma^2)$$

[23]

Consider an option in a market. At any given time t , its price either rises or falls. The evolution of the option's price can thus be modeled as a random walk. According to **Donsker's theorem**, this random walk converges in distribution to a standard Brownian motion $W = (W_t, t \geq 0)$. Consequently, studying the random and chaotic nature of the option's price evolution amounts to studying a Brownian motion.

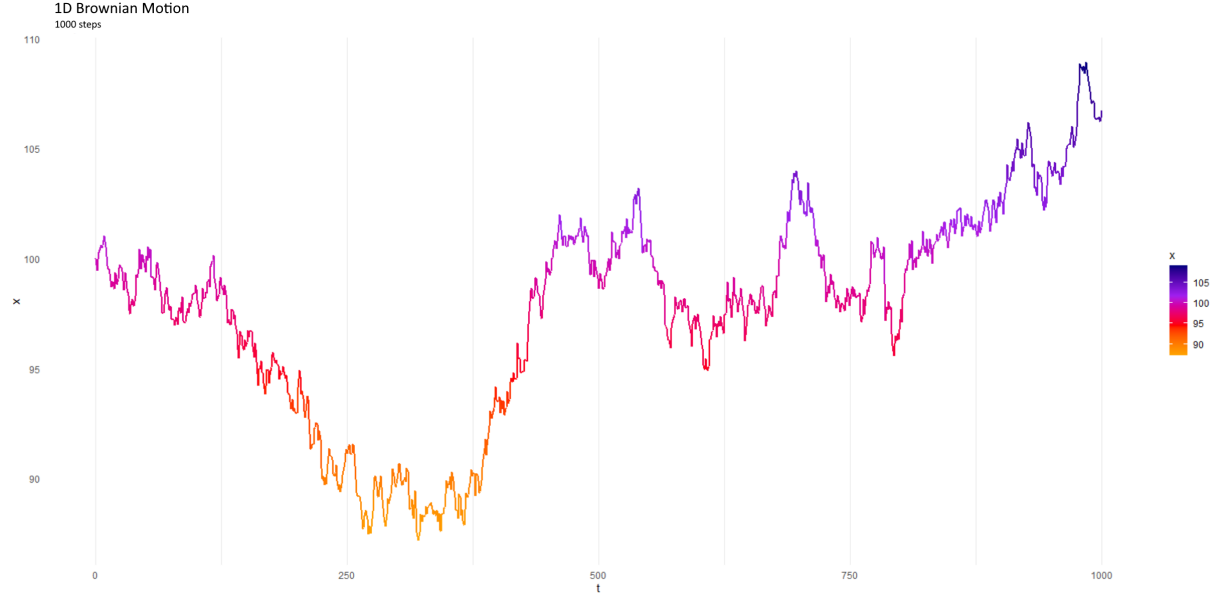


Figure 3: 1D Brownian motion simulation

5.2 Itô Integration

As previously seen, there is no derivative in the classical sense of the term. The mathematician Kiyoshi Itô therefore developed a new theory of integration based on the random nature of the functions under study. We will now focus on the calculation of Itô integrals. These integrals allow for the integration of random variables. The integral is itself a random variable and in particular, is a martingale.

Consider a Wiener process W_t on an interval $I = [0, T]$ with $T \in \mathbb{R}^+$, and a regular partition of this interval $0 = t_0 < t_1 < \dots < t_n = T$, where $n \in \mathbb{N}$. Analogous to Riemann integration, the value of the integral can be approximated using partial sums of rectangles with width $t_{i+1} - t_i$ and height W_{t_i} . The left side of the rectangle must coincide with the value of the integrand at the left endpoint because Brownian motion is a martingale and must not anticipate the future (using the midpoint of each partition segment would yield a Stratonovich integral).

Thus, the Itô integral is defined as the mean-square limit of the Riemann sum evaluated at the left endpoint:

$$\begin{aligned}
I(T) &= \int_0^T W_t dW_t = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} W_{t_i} (W_{t_{i+1}} - W_{t_i}) \\
&\text{using the identity } a(b-a) = \frac{1}{2}(b^2 - a^2 - (b-a)^2) \text{ with } a = W_{t_i} \text{ and } b = W_{t_{i+1}} \\
&\implies W_{t_i} (W_{t_{i+1}} - W_{t_i}) = \frac{1}{2}(W_{t_{i+1}}^2 - W_{t_i}^2) - \frac{1}{2}(W_{t_{i+1}} - W_{t_i})^2 \\
&\implies \sum_{i=0}^{n-1} W_{t_i} (W_{t_{i+1}} - W_{t_i}) = \frac{1}{2} \sum_{i=0}^{n-1} (W_{t_{i+1}}^2 - W_{t_i}^2) - \frac{1}{2} \sum_{i=0}^{n-1} (W_{t_{i+1}} - W_{t_i})^2 \\
&\text{or } \sum_{i=0}^{n-1} (W_{t_{i+1}}^2 - W_{t_i}^2) = W_{t_n}^2 - W_{t_0}^2 = W_T^2 \text{ by telescopic sum} \\
&\text{and } \sum_{i=0}^{n-1} (W_{t_{i+1}} - W_{t_i})^2 \xrightarrow{L^2} T \text{ char } \mathbb{E}[(W_{t_{i+1}} - W_{t_i})^2] = \text{Var}(W_{t_{i+1}} - W_{t_i}) = t_{i+1} - t_i \\
&\implies I(T) = \int_0^T W_t dW_t = \frac{1}{2} W_T^2 - \frac{1}{2} T
\end{aligned}$$

Let us conclude our introduction to this type of integration with a lemma known as **Itô's Lemma**. To do so, we must define the **Itô process**:

$$I_t = \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s \quad \text{which is equivalent to} \quad dI_t = \mu_t dt + \sigma_t dW_t$$

[24] where W_t is a Wiener process, and $\mu \in L^1$ and $\sigma \in L^2$ are deterministic processes.

Remark. The Wiener process has zero expectation, so $\mathbb{E}[I_t] = \int_0^t \mu_s ds$, and non-zero variance, so $\text{Var}[X_t] = \int_0^t \sigma_s^2 ds$

Itô's Lemma

Let $\mu \in L^1$, $\sigma \in L^2$, I_t be an Itô process, and $f(I_t, t)$ be a function of class $\mathcal{C}^2(\mathbb{R} \times \mathbb{R}^+, \mathbb{R})$; then:

$$d(f(I_t, t)) = \frac{\partial f}{\partial t}(I_t, t) dt + \frac{\partial f}{\partial x}(I_t, t) dI_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(I_t, t) \sigma_t^2 dt$$

We previously compared the evolution of an option in a market to Brownian motion. However, an option price cannot be negative. To address this issue, one can define the **log-normal process** or **geometric Brownian motion**:

$$X_t = X_0 e^{(a - \frac{1}{2}\sigma^2)t + \sigma W_t} \quad \text{where } a, \sigma \in \mathbb{R}$$

Thus, geometric Brownian motion is a solution to the stochastic differential equation: $dX_t = aX_t dt + \sigma X_t dW_t$

Proof Let $X_t = X_0 e^{(a - \frac{1}{2}\sigma^2)t + \sigma W_t}$ where $a, \sigma \in \mathbb{R}$. We then define the Itô process $I_t = (a - \frac{1}{2}\sigma^2)t + \sigma W_t$. We then have $dI_t = \underbrace{(a - \frac{1}{2}\sigma^2)}_{\mu_t} dt + \underbrace{\sigma}_{\sigma_t} dW_t$. We define the function $f(x, t) = X_0 e^x$

such that $f(I_t, t) = X_t$

- $\frac{\partial f}{\partial t}(I_t, t) = 0$
- $\frac{\partial f}{\partial x}(I_t, t) = X_0 e^{I_t} = X_t$
- $\frac{\partial^2 f}{\partial x^2}(I_t, t) = X_0 e^{I_t} = X_t$

We can then apply Itô's lemma:

$$dX_t = 0 \cdot dt + X_t dI_t + \frac{1}{2} X_t \sigma^2 dt$$

Let us substitute dI_t by $(a - \frac{1}{2}\sigma^2)dt + \sigma dW_t$:

$$\begin{aligned} dX_t &= X_t \left[\left(a - \frac{1}{2}\sigma^2 \right) dt + \sigma dW_t \right] + \frac{1}{2} X_t \sigma^2 dt \\ &= aX_t dt - \frac{1}{2}\sigma^2 X_t dt + \sigma X_t dW_t + \frac{1}{2} X_t \sigma^2 dt \\ &= aX_t dt + \sigma X_t dW_t \end{aligned}$$

Finally, we have:

$$dX_t = aX_t dt + \sigma X_t dW_t$$

After having defined all these processes, we will be able to apply them to the Black-Scholes model in the following subsection.

5.3 Black-Scholes Equation

Let us now introduce the most important partial differential equation in financial mathematics: the Black-Scholes equation, named after Fischer Black and Myron Scholes, who presented it in an academic research paper. We will first examine the Black-Scholes model before demonstrating that, as time steps are reduced, the Cox-Ross-Rubinstein model converges to the Black-Scholes model.

Black-Scholes-Merton Partial Differential Equation

Let $V(S, t)$ be the price of an option, where S is the price of the underlying asset and t represents time:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

[25]

The model relies on a large number of rather restrictive assumptions, which are necessary for the derivation of the formula. Since the objective here is to study the Cox-Ross-Rubinstein model, I will present only the simplified assumptions needed to understand the solution. However, the full set of assumptions can be found on the Wikipedia page for the Black-Scholes model.

The assumptions of this model are therefore:

- The risk-free rate is constant
- The price follows geometric Brownian motion and the variance of the motion is constant
- The option pays no dividends
- The market satisfies the no-arbitrage condition
- The market allows the sale of an asset in any quantity
- Transactions incur no costs

The Black-Scholes formula allows for the calculation of the price of a European call or put but does not allow for the calculation of American calls or puts, unlike the Cox-Ross-Rubinstein model.

TODO [26]

5.4 Convergence

TODO

6 Risk indicators

In this section, we will first examine the theory surrounding risk indicators known as **Greeks**, and then use the binomial model to approximate them. We will limit our scope to the primary Greeks: $\Delta, \Gamma, \nu, \Theta$. While many others exist—such as ρ, λ, ϵ —they are less relevant given the assumptions underlying the binomial model (constant risk-free rate, no dividends, etc.).

TODO

7 Comparative analysis of others models

8 Conclusion

TODO

ANNEXES

Code du Modèle CRR (C++)

Construction de l'arbre binomial

```
1      Node * Node::createTree(int tree_depth) {
2          if (tree_depth <= 0) {
3              return nullptr;
4          }
5          std::queue<Node *> previous_level_queue;
6
7          Node * root = new Node(0,0);
8          previous_level_queue.push(root);
9
10         for(int i = 1; i < tree_depth; i++) {
11             std::queue<Node *> current_level_queue;
12
13             for (int j = 0; j <= i; j++) {
14                 Node * node = new Node(i * (i + 1) / 2 + j, i);
15                 node->setValue(0);
16                 current_level_queue.push(node);
17             }
18
19             std::queue<Node *> temp_children = current_level_queue;
20
21             while (!previous_level_queue.empty()) {
22                 Node * parent = previous_level_queue.front();
23                 previous_level_queue.pop();
24
25                 Node * downChild = temp_children.front();
26                 parent->setDown(downChild);
27                 downChild->setPrevUp(parent);
28
29                 temp_children.pop();
30
31                 Node * upChild = temp_children.front();
32                 parent->setUp(upChild);
33                 upChild->setPrevDown(parent);
34             }
35             previous_level_queue = current_level_queue;
36         }
37         return root;
38     }
```

Évaluation des payoffs

```

1      void CRR::evaluateLeafNodes(Node * t_price, Node * t_premium, int
2      type, double strike) {
3          Node * iter = t_price;
4          Node * iter2 = t_premium;
5
6          while(iter->down() != nullptr) {
7              iter = iter->down();
8              iter2 = iter2->down();
9          }
10
11         while(iter != nullptr) {
12             if (type == 0)
13                 iter2->updateCallPrice(iter, strike);
14             else
15                 iter2->updatePutPrice(iter, strike);
16
17             if (iter->prevUp() != nullptr) {
18                 iter = iter->prevUp()->up();
19                 iter2 = iter2->prevUp()->up();
20             } else {
21                 break;
22             }
23         }
24     }

```

Backward induction

```
1  void CRR::backwardInduction(Node * node, double u, double d, double r
2  , double time_value, int depth) {
3      double dt = time_value / depth;
4      double R = exp(r * dt);
5      double p = (R - d)/(u - d);
6      double q = 1. - p;
7      Node * iter = node;
8
9      while(iter->down()) {
10         iter = iter->down();
11     }
12
13     if(!iter->prevUp()) return;
14     iter = iter->prevUp();
15
16     while(iter->prevUp()) {
17         Node * parent = iter->prevUp();
18         while(iter->up() && iter->up()->prevUp()) {
19             iter->setValue(CRR::getNodeMeanValue(iter, R, p, q));
20             iter = iter->up()->prevUp();
21         }
22         iter->setValue(CRR::getNodeMeanValue(iter, R, p, q));
23         iter = parent;
24     }
25     iter->setValue(CRR::getNodeMeanValue(iter, R, p, q));
26 }
```

Simulations (R)

Simulation d'un mouvement brownien 2D

```
1  library(ggplot2)
2
3  n <- 2500
4  dt <- 0.01
5
6  df <- data.frame(
7      x = cumsum(c(0, rnorm(n, 0, sqrt(dt)))),
8      y = cumsum(c(0, rnorm(n, 0, sqrt(dt)))),
9      t = 0:n
10 )
11
12  ggplot(df, aes(x, y, color = t)) + geom_path() +
13      scale_color_gradientn(colors = c("orange", "red", "purple", "navy
14      ")) + coord_fixed() + labs(title = "Mouvement Brownien 2D",
15      subtitle = paste(n, "pas"))
```

Simulation d'un mouvement brownien 1D

```
1   library(ggplot2)
2
3   n <- 1000
4   dt <- 0.0001
5
6   df <- data.frame(
7     x = cumsum(c(100, rnorm(n, 0, 0.5))),
8     t = 0:n
9   )
10
11  ggplot(df, aes(t, x, color = x)) + geom_line(linewidth = 1) +
12  scale_color_gradientn(colors = c("orange", "red", "purple", "navy
    ")) + coord_cartesian(xlim = c(0, n)) + theme_minimal() +
    theme(panel.grid.major.y = element_blank(), panel.grid.minor.
      y = element_blank()) + labs(title = "Mouvement_Brownien_1D",
        subtitle = paste(n, "pas"))
```

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